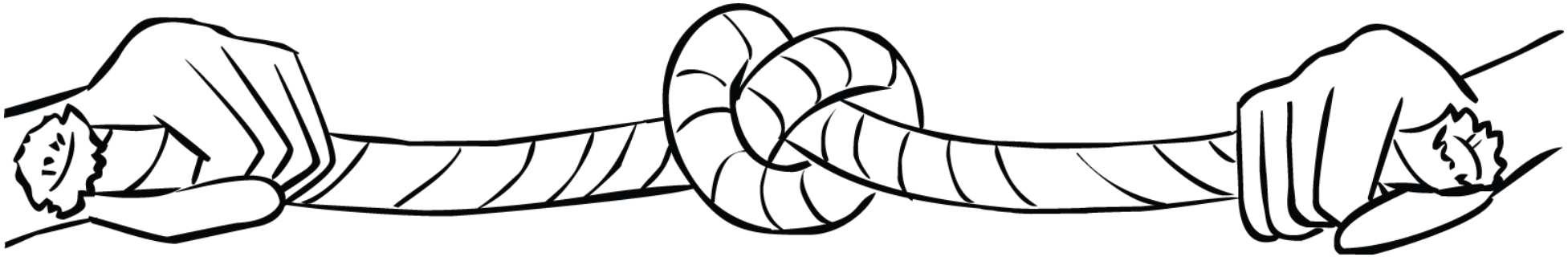
2. Knots

If a peace of rope is attached to the end of another rope with a knot, the knot may fail to hold, break or untie under certain conditions. Investigate factors that affect the knot strength.

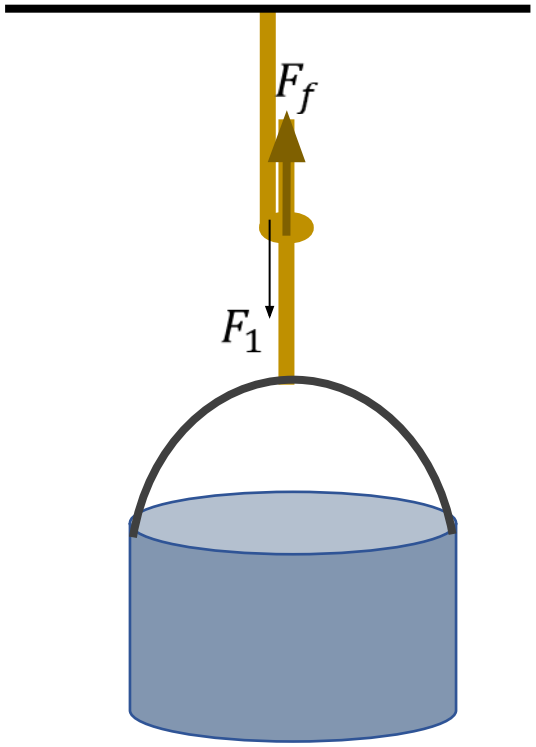


Plan

- Explanation of the phenomenon
  - *The rope doesn't slip;*
  - *The rope slips;*
- Theoretical model
  - *Vector quality of the forces;*
- Experiments
  - *Experimental setup;*
  - *Different knot tension;*
  - *Different coefficient of friction of the rope;*
  - *Number of knots;*
- Comparison of theoretical and experimental results
  - *Theoretical and experimental graphs;*
- Conclusion
  - *Important parameters;*
  - *Analysis of results;*

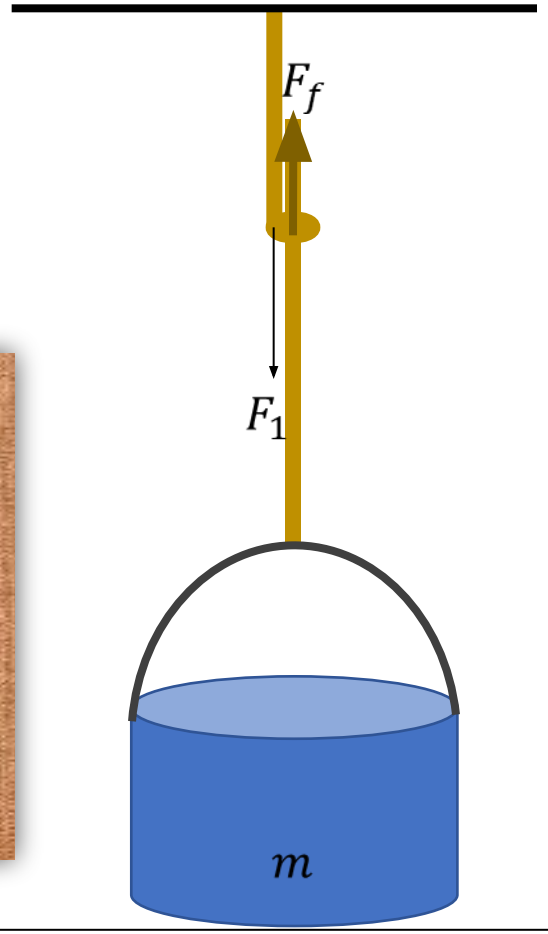
Explanation of the phenomenon

Phenomenon



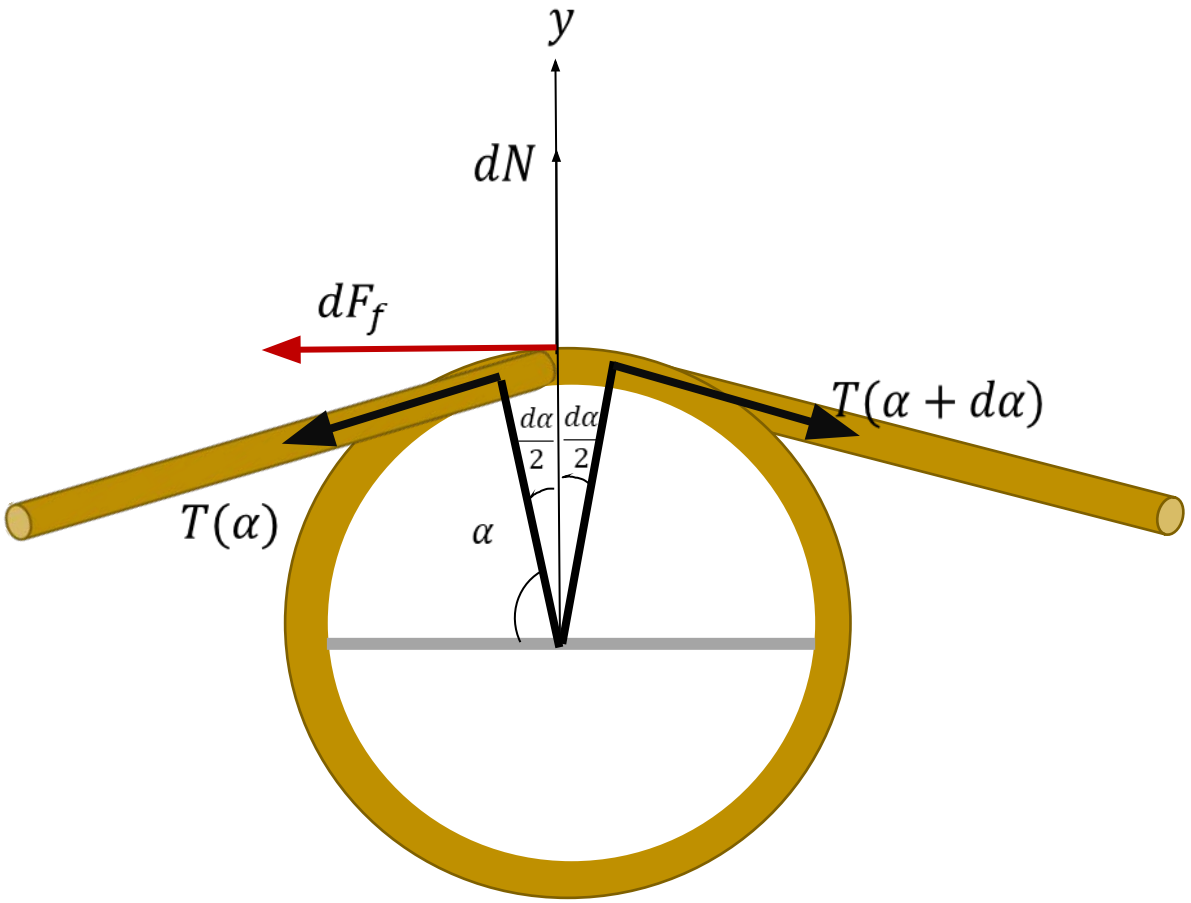
Slip Condition

$$F > F_f$$



Theoretical model

Determination of critical force



$$y: \quad dN = T(\alpha)sin\frac{d\alpha}{2} + T(\alpha + d\alpha)sin\frac{d\alpha}{2}$$

$$sin\frac{d\alpha}{2} \approx \frac{d\alpha}{2}$$

$$dN = \frac{d\alpha}{2}\big(T(\alpha) + T(\alpha + d\alpha)\big)$$

$$T(\alpha) + T(\alpha + d\alpha) \approx 2T(\alpha)$$

$$dN = d\alpha T(\alpha)$$

$$\boxed{dN = d\alpha T(\alpha)}$$

Determination of critical force

x: $T(\alpha + d\alpha) = \mu dN + T(\alpha)$

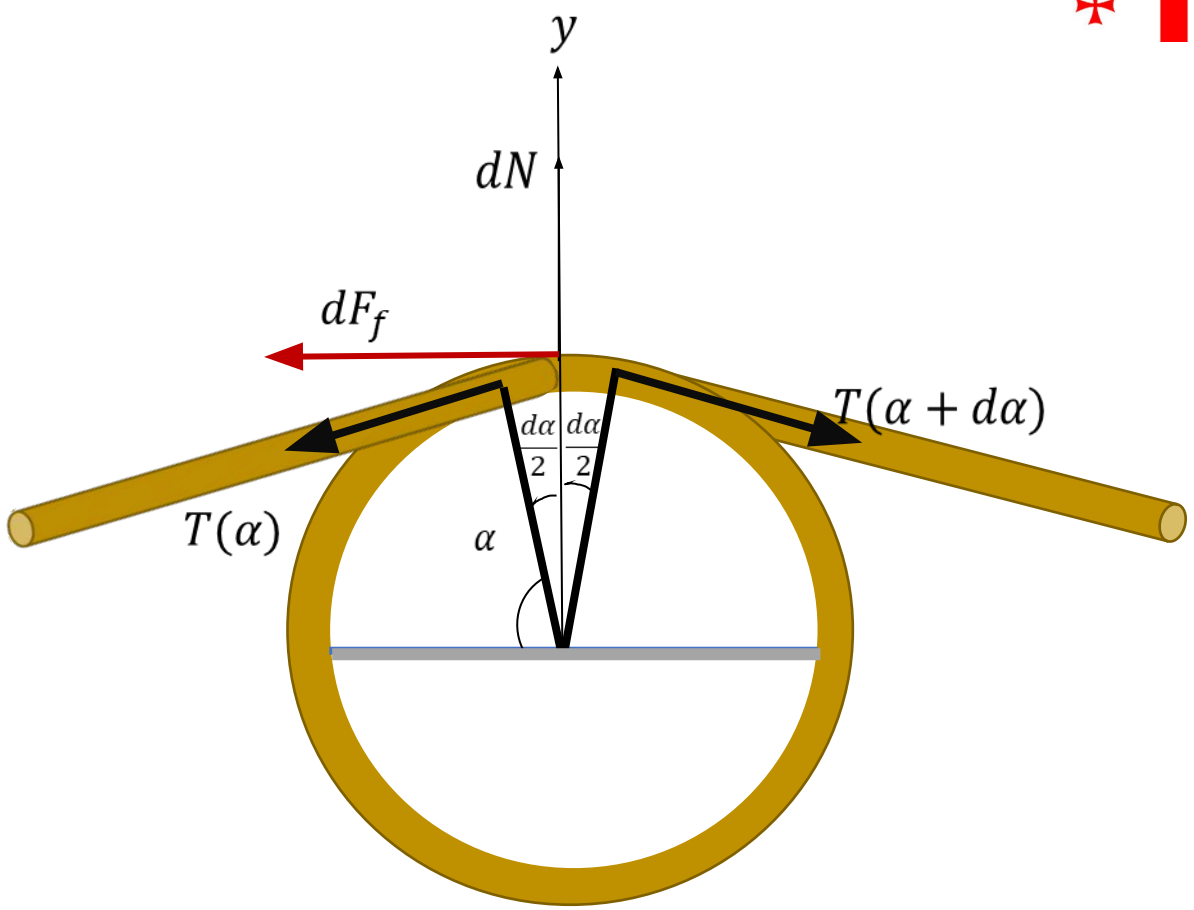
$$T(\alpha + d\alpha) - T(\alpha) = \mu d\alpha T(\alpha)$$

$$\frac{T(\alpha + d\alpha) - T(\alpha)}{d\alpha} = \mu T(\alpha)$$

$$T'(\alpha) = \mu T(\alpha)$$

$$T(\alpha) = ce^{\mu\alpha}$$

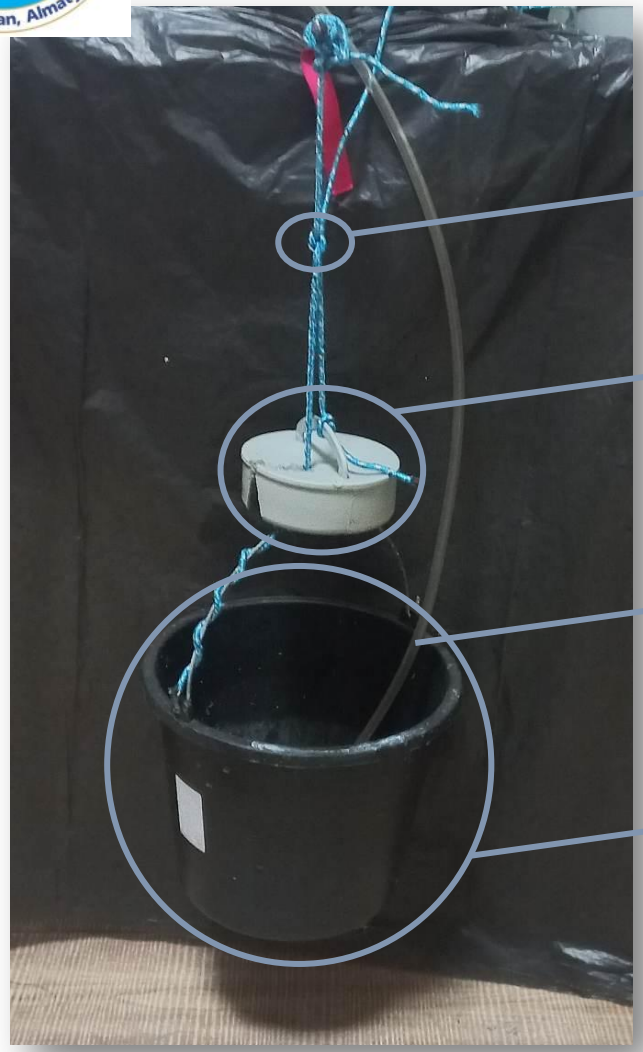
$$\boxed{T = ce^{\mu\alpha}}$$



$$\boxed{T_{max} = \mu N + ce^{\mu\alpha}}$$

Experimental part

Experimental setup and variable parameters



Knot

Constant mass

Water tube

Variable mass

Constant mass(kg)
0,33kg
0,5kg
1kg
1,5kg
2kg

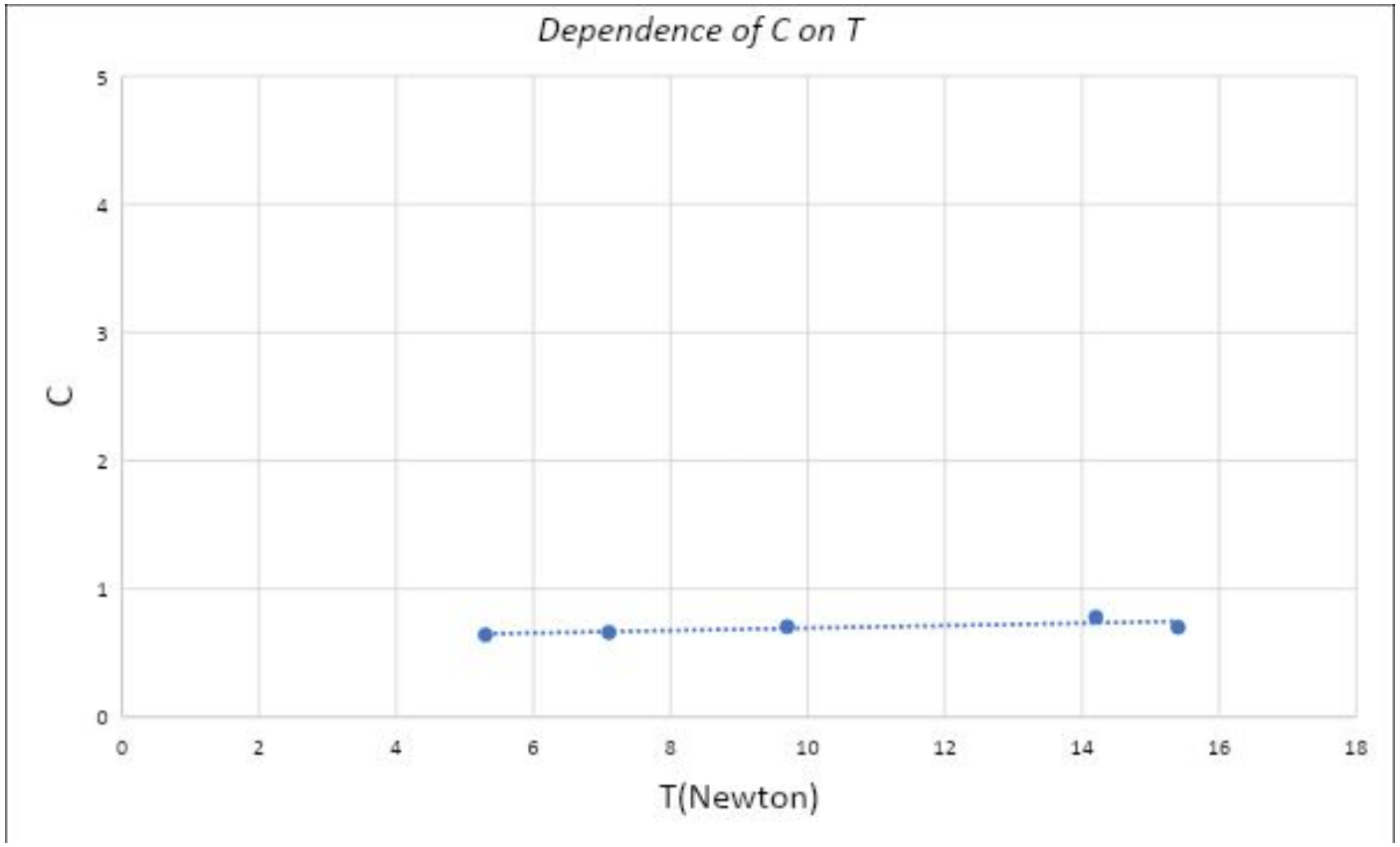
Friction coefficient
0,38
0,57
0,62
0,75

Checking the constancy of the C coefficient

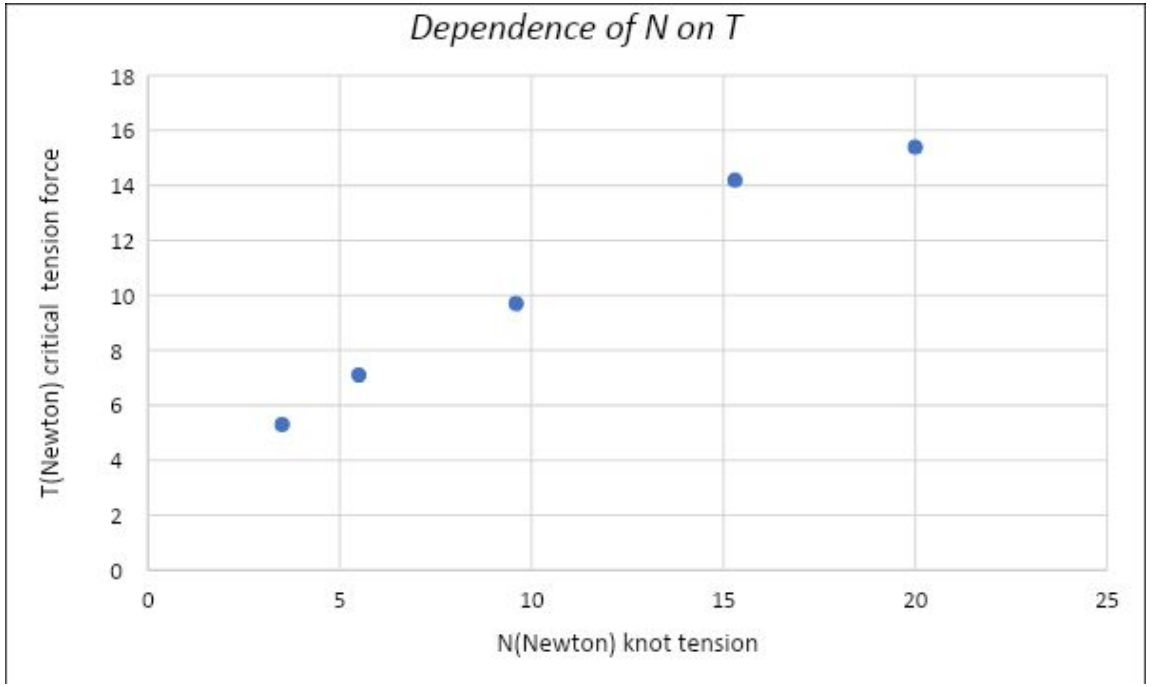
$$C = \frac{T - \mu N}{e^{\mu\alpha}}$$

C is constant

$C \approx 0.7$

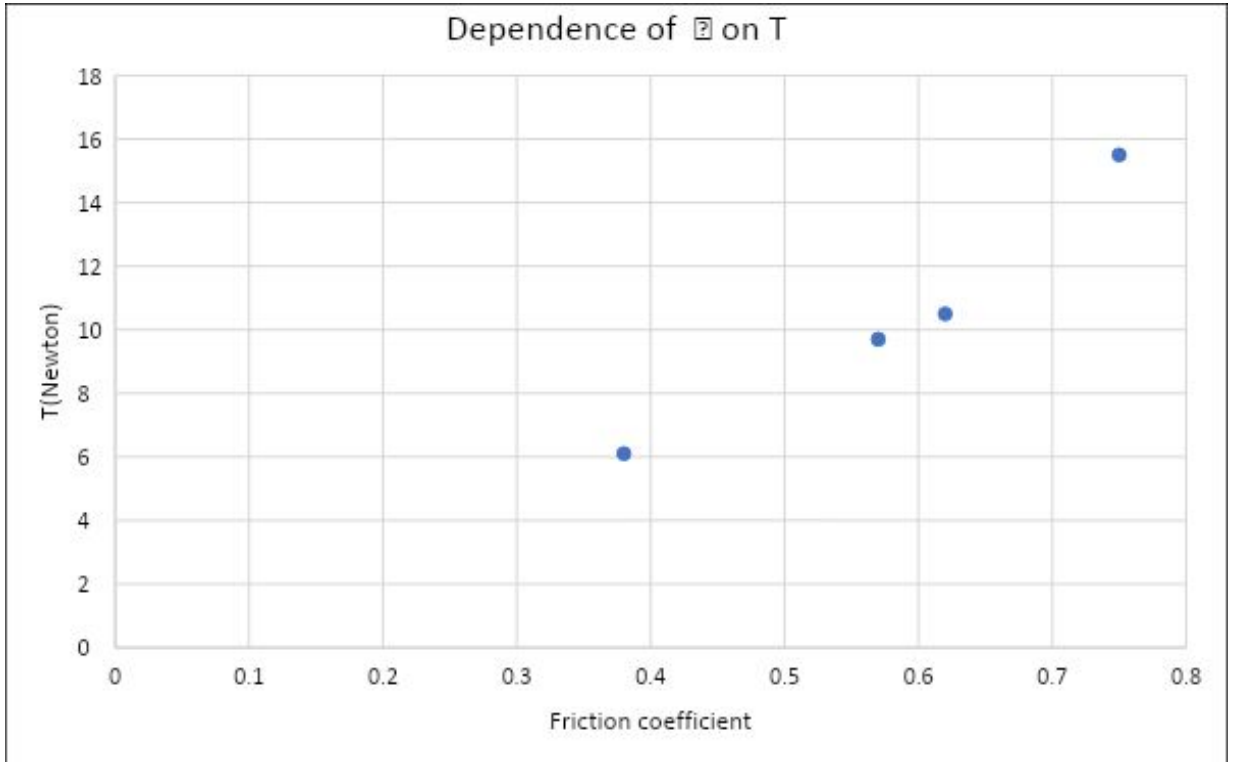


Different knot tension



$$c \approx 0.7; \mu = 0.57$$

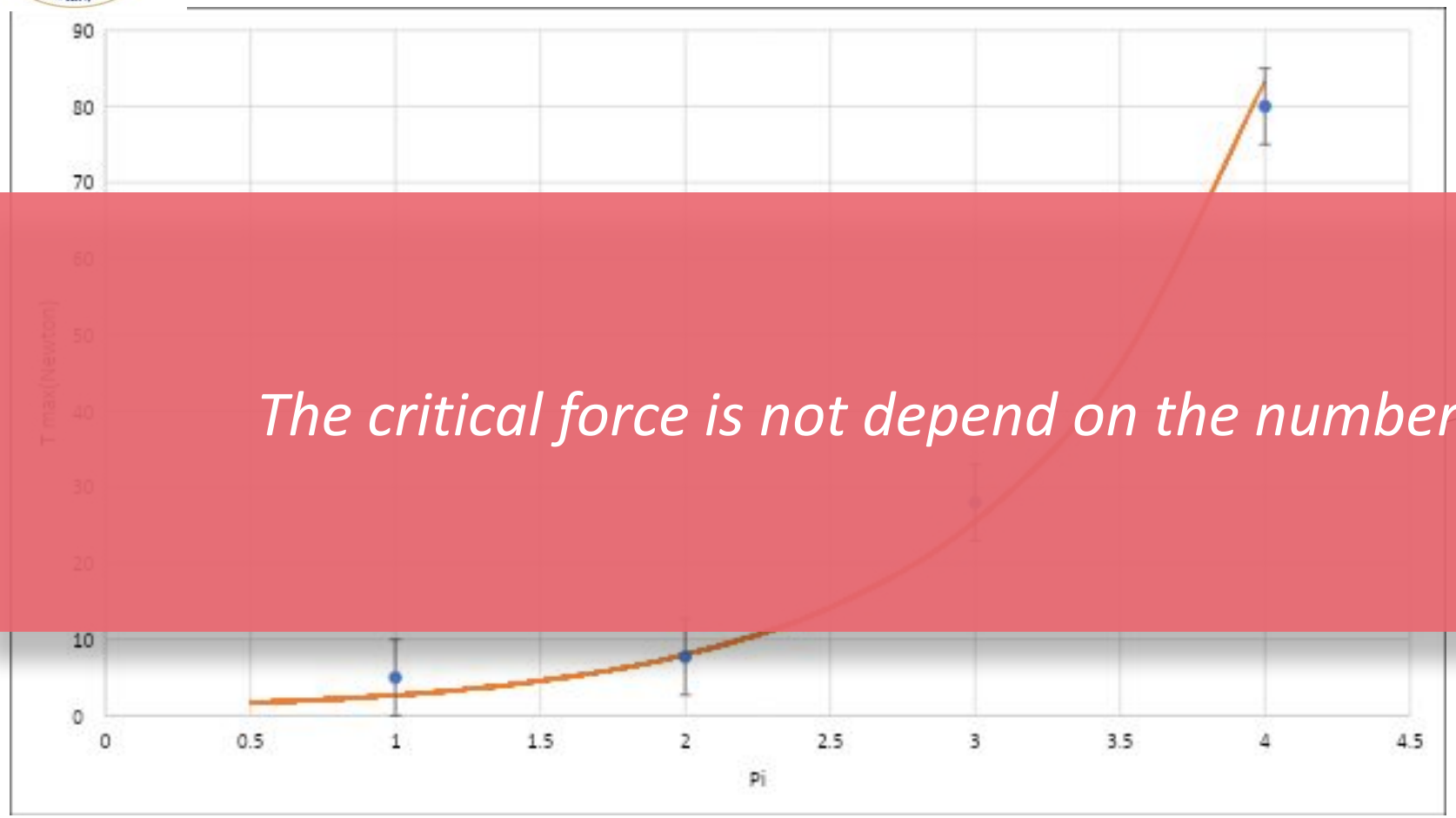
Different friction coefficient

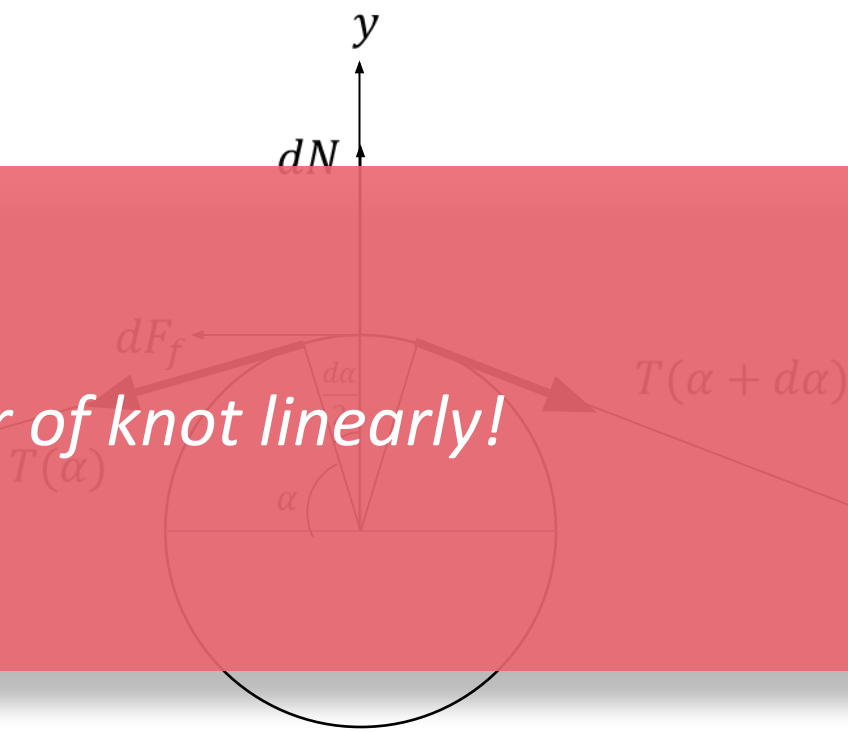


$$tg\beta = \mu$$

$c \approx 0.7$

Number of coils of the knots

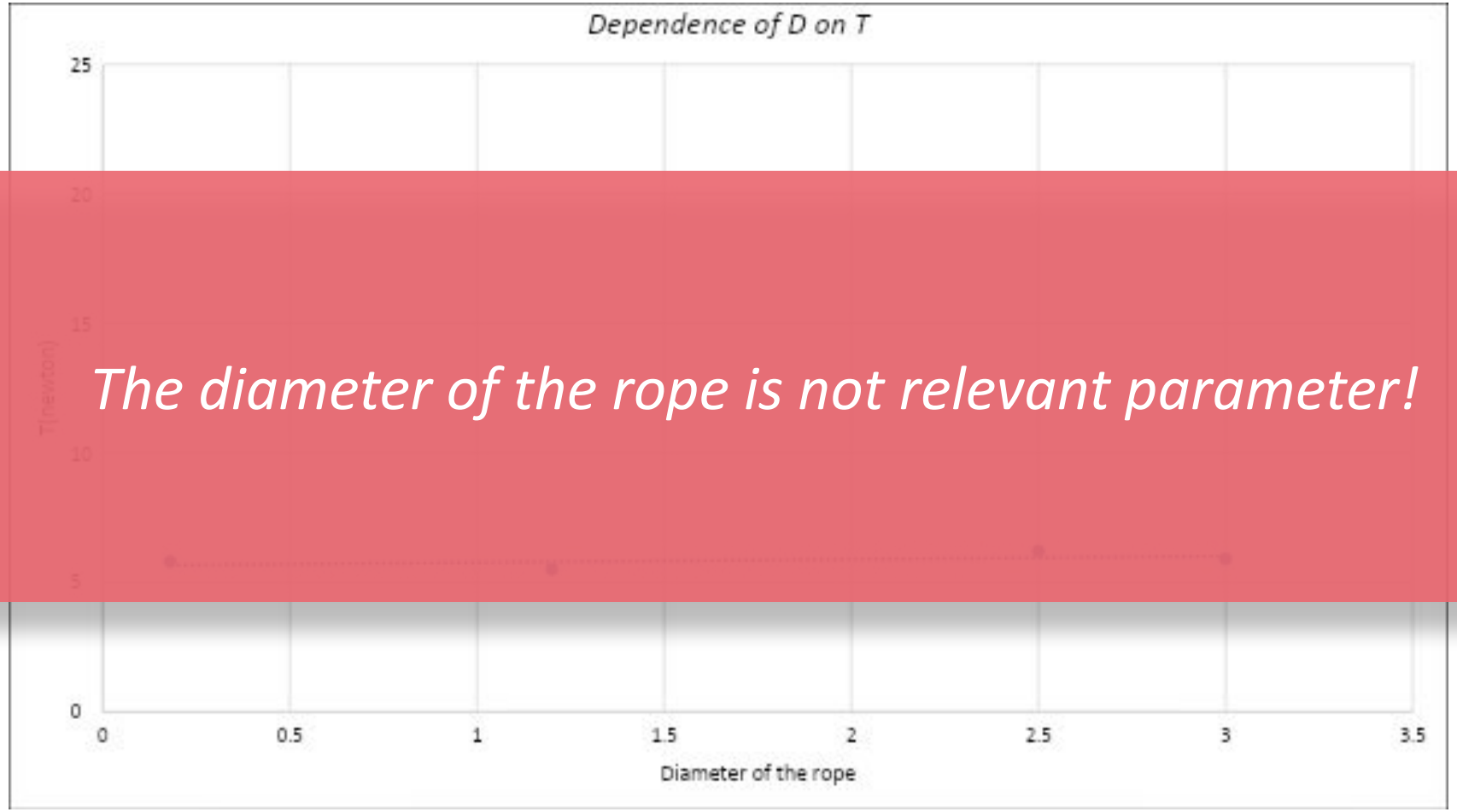




The critical force is not depend on the number of knot linearly!

phenomenon | Theoretical model | experiment | conclusion

Different diameters of the rope

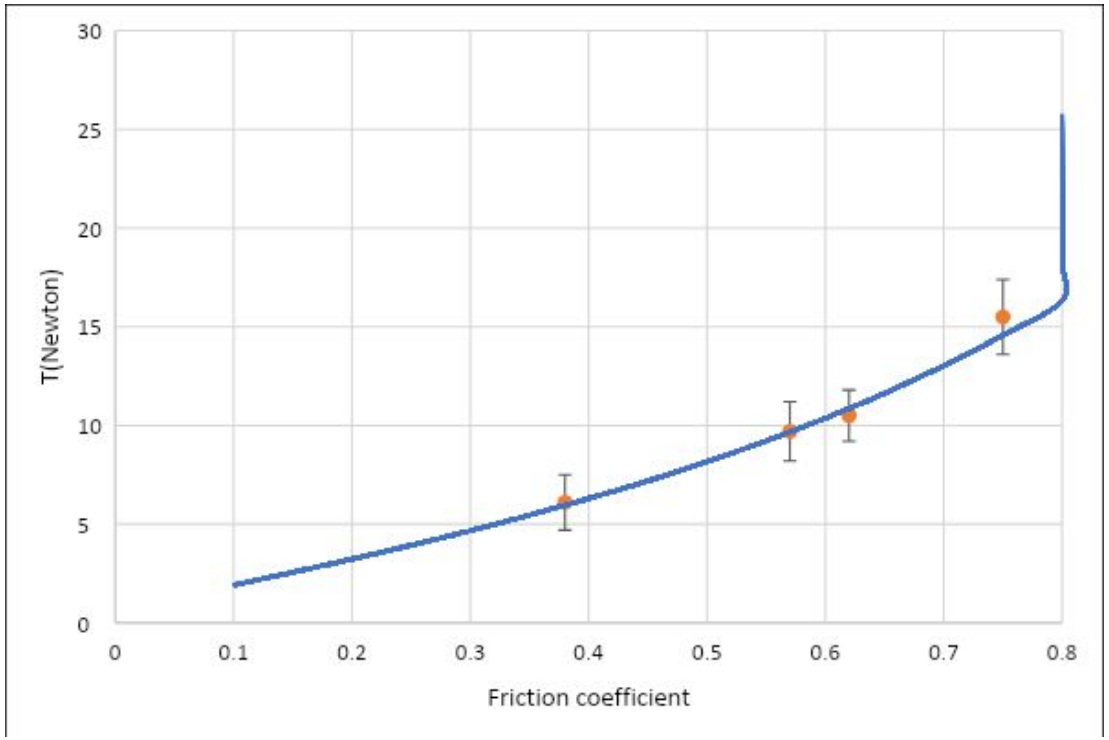


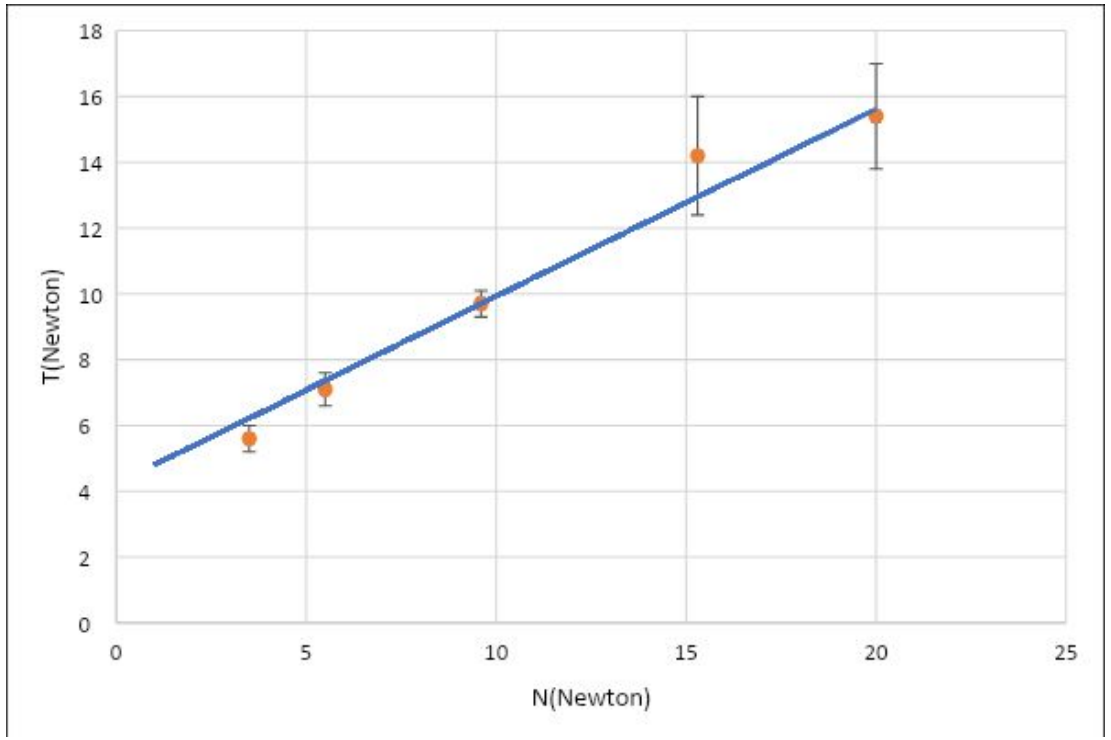
The diameter of the rope is not relevant parameter!

The final part

Theoretical and experimental graphs

- Experimental
- Theoretical

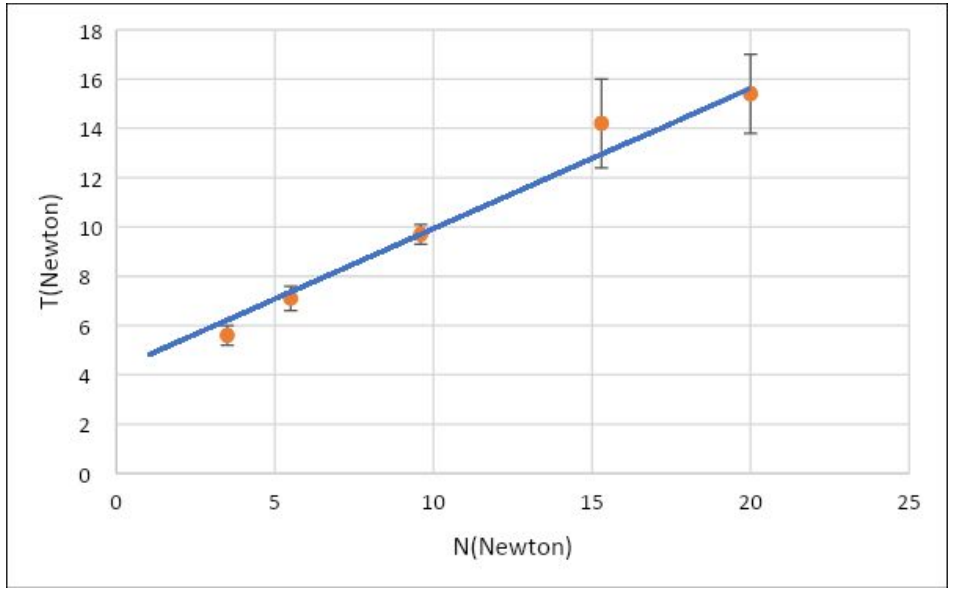


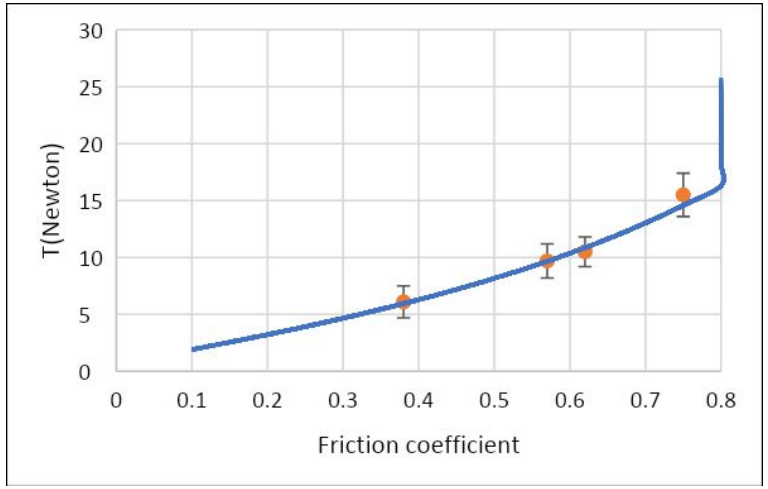


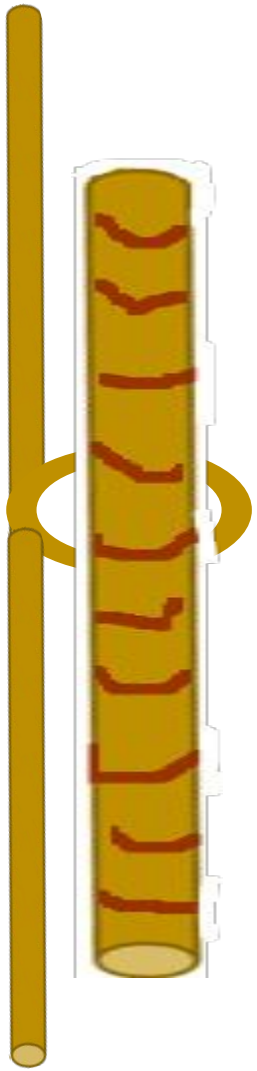
Relevant parameters

- Knot tension;
- Friction coefficient;
- The angle of contact of the rope;
- C coefficient;

$$C \approx 0.7$$





$$T = \mu N + ce^{\mu\alpha}$$

Thanks for attention!